On the Submatrices with the Best-Bounded Inverses for Matrices with Three Orthonormal Columns

Tongren Xiao

Abstract

Let $U \in \mathbb{R}^{n\times 3}$ have orthonormal columns. We prove a lower bound $\Lambda_n \geq \tau_n$ for the maximum of the squared smallest singular values of its three-row submatrices, and obtain

$$\liminf_{n\to\infty} n\Lambda_n \geq \frac{5-\sqrt{13}}{2}.$$

We prove the bound $n^{-1/2}$ when the nonzero rows of U belong to at most five one-dimensional linear subspaces of $\mathbb{R}^3$, and give a sharpness construction showing that this constant is optimal. More generally, the constant $n^{-1/2}$ cannot be increased for any $n > k$. In particular, $\Lambda_5 = 1/5$. For six rows, we prove the lower bound α_6, where α_6 is the smallest zero of $70x^3 - 120x^2 + 57x - 6$, and show that every possible counterexample must have six distinct row directions, two complementary three-row bases, and projection diagonal in a bounded region.

1 Introduction

Let $U \in \mathbb{R}^{n\times k}$ satisfy $U^TU = I_k$. Goreinov, Tyrtyshnikov, and Zamarashkin conjectured that there is a set $I \subseteq \{1,\ldots,n\}$ of cardinality k such that

$$\sigma_{\min}(U_I) \geq \frac{1}{\sqrt{n}}, \tag{1}$$

where U_I is formed by the rows of U indexed by I [1]. The cases $k = 1$ and $k = n-1$ are elementary. Nesterenko proved $(n,k) = (4,2)$ [3], and Sengupta and Pautov proved the real case $k = 2$ for arbitrary n [5]. We consider real matrices with $k = 3$.

For $n \geq 4$, define

$$\Lambda_n = \inf_{\substack{U\in\mathbb{R}^{n\times 3}\\ U^TU=I_3}} \max_{|I|=3} \sigma_{\min}(U_I)^2.$$

For $n \geq 6$, put

$$\tau_n = \frac{5n^2 - 11n + 4 - \sqrt{(n^2-3n+4)(13n^2-43n+36)}}{2(n-2)(n^2+n-4)}.$$

Theorem 1.1. *For every $n \geq 6$,*

$$\Lambda_n \geq \tau_n > \frac{2}{3n-4}.$$

Theorem 1.2. *One has*

$$\liminf_{n\to\infty} n\Lambda_n \geq \frac{5-\sqrt{13}}{2}. \tag{2}$$

The next theorem proves the the constant under a restriction on the row directions. A line means a one-dimensional linear subspace of $\mathbb{R}^3$.

Theorem 1.3. *Let $n > 3$ and let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^T U = I_3$. If every nonzero row of U belongs to one of at most five lines, then there is a three-element set I such that*

$$\sigma_{\min}(U_I) \geq \frac{1}{\sqrt{n}}.$$

Theorem 1.4. *For every $n \geq 4$, there is a matrix $U \in \mathbb{R}^{n\times 3}$ with $U^T U = I_3$ such that*

$$\max_{|I|=3} \sigma_{\min}(U_I)^2 = \frac{1}{n}.$$

Its nonzero rows lie on four lines.

Theorem 1.5. *Let $n > k \geq 1$. There is a matrix $V \in \mathbb{R}^{n\times k}$ with $V^T V = I_k$ such that*

$$\max_{|J|=k} \sigma_{\min}(V_J)^2 = \frac{1}{n}. \tag{3}$$

For $k \geq 2$, the nonzero rows of V lie on $k+1$ lines. For $k = 1$, take the matrix whose rows are all equal to $n^{-1/2}$.

Corollary 1.6. $\Lambda_5 = 1/5$.

Proof. Theorem 1.3 gives $\Lambda_5 \geq 1/5$, while Theorem 1.4 gives the reverse inequality. $\square$

For six rows, let α_6 be the smallest zero of

$$70x^3 - 120x^2 + 57x - 6.$$

Theorem 1.7. *If $U \in \mathbb{R}^{6\times 3}$ and $U^T U = I_3$, then some three-element set I satisfies*

$$\sigma_{\min}(U_I)^2 \geq \alpha_6 > \frac{1}{7}.$$

Theorem 1.8. *Suppose that $U \in \mathbb{R}^{6\times 3}$, $U^T U = I_3$, and*

$$\max_{|I|=3} \sigma_{\min}(U_I)^2 < \frac{1}{6}. \tag{4}$$

Put $P = UU^T$. Then the six nonzero rows of U determine six distinct lines, there is a three-element set I for which both U_I and U_{I^c} are nonsingular, and

$$\frac{5}{36} < P_{ii} < \frac{31}{36} \quad (1 \leq i \leq 6), \qquad \sum_{i=1}^{6} \left(P_{ii} - \frac{1}{2}\right)^2 < \frac{8}{27}.$$

Moreover,

$$\alpha_6 \leq \max_{|I|=3} \sigma_{\min}(U_I)^2 < \frac{1}{6}.$$

Section 2 establishes the projection identities, the mixed characteristic polynomial, and the inequalities for the quantities Q, B, C used throughout the paper. Section 3 proves Theorems 1.1 and 1.2. Section 4 proves Theorems 1.3 and 1.4. Section 5 proves Theorems 1.7 and 1.8. Section 6 summarizes the results and states the remaining problem.

2 Preliminaries

For a real symmetric matrix A, the notation $A \succeq 0$ means that A is positive semidefinite, and $A \succeq B$ means $A - B \succeq 0$. We write $e_r(A)$ for the rth elementary symmetric function of the eigenvalues of A. Thus $e_r(A)$ is the sum of all $r \times r$ principal minors of A.

Fact 2.1. *If $U \in \mathbb{R}^{n\times 3}$ has orthonormal columns, then*

$$P = UU^T$$

is the orthogonal projection onto the column space of U. For a set $I \subseteq \{1, \dots, n\}$, the matrices $P_{I,I}$ and $U_I U_I^T$ are equal. Consequently,

$$\sigma_{\min}(U_I)^2 = \lambda_{\min}(P_{I,I}) \tag{5}$$

when $|I| = 3$.

Fact 2.2. *Let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^T U = I_3$, put $P = UU^T$, and define*

$$d_I = \det(P_{I,I}) \quad (|I| = 3), \qquad b_{ij} = \det(P_{\{i,j\},\{i,j\}}).$$

Then

$$\sum_{|I|=3} d_I = 1.$$

$$\sum_{I \ni i} d_I = P_{ii}.$$

$$\sum_{I \supset \{i,j\}} d_I = b_{ij}.$$

Proof. The first identity is the Cauchy–Binet formula applied to $U^T U$. For a fixed set J of one or two indices, apply Cauchy–Binet after decomposing the exterior product of the columns of U according to whether its row index contains J. Equivalently, differentiate $\det(U^T DU)$ with respect to the diagonal entries of D and set $D = I_n$. This gives

$$\sum_{I \supset J} \det(P_{I,I}) = \det(P_{J,J}),$$

which yields the two displayed marginal identities. □

Definition 2.3. A real two-form $q = (q_{ij})_{1\le i<j\le m}$ is called decomposable if there exist $x, y \in \mathbb{R}^m$ such that

$$q_{ij} = x_i y_j - x_j y_i \qquad (1 \le i < j \le m).$$

We use the following Schur-complement criterion.

Fact 2.4. *Let $A \succ 0$ and let X be a real matrix with the same number of rows as A. Then*

$$A - XX^T \succeq 0 \quad \iff \quad X^T A^{-1} X \preceq I.$$

Proof. Congruence by $A^{-1/2}$ reduces the assertion to $I - YY^T \succeq 0$ if and only if $Y^T Y \preceq I$, where $Y = A^{-1/2}X$. The nonzero eigenvalues of YY^T and $Y^T Y$ coincide. □

2.1 Principal submatrices

For a monic real-rooted polynomial p, write $\lambda_{\min}(p)$ for its smallest zero.

Fact 2.5. *Let $A \in \mathbb{R}^{n\times n}$ be symmetric, let $p_A(x) = \det(xI_n - A)$, and let $1 \le k \le n$. Then*

$$\binom{n}{k}^{-1} \sum_{|I|=k} \det(xI_k - A_{I,I}) = \frac{p_A^{(n-k)}(x)}{\binom{n}{k}(n-k)!}. \tag{6}$$

Moreover, there is a k-element set I such that

$$\lambda_{\min}(A_{I,I}) \ge \lambda_{\min}\left(\frac{p_A^{(n-k)}}{\binom{n}{k}(n-k)!}\right). \tag{7}$$

Proof. Repeated differentiation of the determinant gives

$$p_A^{(n-k)}(x) = (n-k)! \sum_{|I|=k} \det(xI_k - A_{I,I}),$$

which proves (6).

We prove the selection statement by recursive deletion. We first record the following interlacing fact. Let B be a real symmetric $m \times m$ matrix and $0 \le r \le m-2$. Then the m polynomials

$$q_i(x) = \frac{d^r}{dx^r} \det(xI_{m-1} - B_{\widehat{i},\widehat{i}}) \qquad (1 \le i \le m) \tag{8}$$

have a common interlacing. Indeed, for nonnegative w_i, not all zero, put $D = \operatorname{diag}(w_1, \dots, w_m)$. If B has simple eigenvalues λ_j with orthonormal eigenvectors v_j, then

$$\begin{aligned} \sum_i w_i \det(xI_{m-1} - B_{\widehat{i},\widehat{i}}) &= \det(xI_m - B) \operatorname{tr}\big(D(xI_m - B)^{-1}\big) \\ &= \det(xI_m - B) \sum_{j=1}^{m} \frac{v_j^T D v_j}{x - \lambda_j}. \end{aligned}$$

All residues in the preceding partial-fraction identity are nonnegative. The polynomial on the left is therefore real-rooted and interlaces p_B; repeated eigenvalues follow by approximation. Its rth derivative is real-rooted by Rolle's theorem. Thus every nonnegative linear combination of the polynomials in (8) is real-rooted. The elementary compatibility criterion for real-rooted polynomials of the same degree and with positive leading coefficients now gives a common interlacing: for two polynomials the assertion follows by alternating signs between successive zeros, and the pairwise inequalities between consecutive zeros yield one common interlacing for the whole finite family.

If monic polynomials q_i have a common interlacing and $q = \sum_i \theta_i q_i$, where $\theta_i \ge 0$ and $\sum_i \theta_i = 1$, then

$$\lambda_{\min}(q_i) \ge \lambda_{\min}(q) \tag{9}$$

for at least one i. Otherwise, at $\lambda_{\min}(q)$ all q_i would have the same sign: before the smallest zero of the common interlacing, each q_i changes sign only at its own smallest zero. Equivalently, the smallest zero of a convex combination lies between the smallest and the largest of the smallest zeros of the q_i. The assertion for coincident zeros follows by continuity.

We apply this observation recursively. For a set S of already deleted coordinates, with $|S| \le n-k$, define

$$p_S(x) = \binom{n-|S|}{k}^{-1} \sum_{\substack{I \subseteq [n]\setminus S \\ |I|=k}} \det(xI_k - A_{I,I}).$$

When $|S| < n-k$, incidence counting gives

$$p_S(x) = \frac{1}{n-|S|} \sum_{i \notin S} p_{S\cup\{i\}}(x).$$

By (6), the children in the preceding averaging identity are positive constant multiples of the same-order derivatives of the polynomials obtained by deleting one coordinate from $A_{[n]\setminus S,[n]\setminus S}$. They therefore have a common interlacing by (8). Equation (9) permits a choice of a child whose smallest zero is no smaller than that of p_S. After $n-k$ choices we reach a k-element set I; the leaf polynomial is $\det(xI_k - A_{I,I})$. This proves (7). □

The following fourth-moment inequality for decomposable two-forms.

Theorem 2.6. *Let $m \geq 3$. If $q = (q_{ij})_{1\leq i<j\leq m}$ is a decomposable real two-form on $\mathbb{R}^m$, then*

$$\sum_{i<j} q_{ij}^4 \geq \frac{3}{4m} \sum_{i=1}^{m} \left(\sum_{j\neq i} q_{ij}^2 \right)^2 \geq \frac{3}{m^2} \left(\sum_{i<j} q_{ij}^2 \right)^2. \tag{10}$$

Both constants are optimal.

Proof. By homogeneity, write $q = x \wedge y$, where $x, y \in \mathbb{R}^m$ are orthonormal. Put

$$P = xx^T + yy^T, \qquad r_i = P_{ii} = \sum_{j\neq i} q_{ij}^2,$$

and let D be the symmetric matrix with zero diagonal and $D_{ij} = q_{ij}^2$ for $i \neq j$. Lagrange's identity gives

$$D = rr^T - P \circ P, \qquad D\mathbf{1} = r,$$

where $\circ$ denotes the entrywise product.

For each i, define

$$w_i = (x_i^2, \sqrt{2}x_i y_i, y_i^2) \in \mathbb{R}^3$$

and let W be the matrix with rows w_i. Then

$$P \circ P = WW^T, \qquad r = W(1,0,1)^T.$$

It follows from the preceding matrix identity that $\operatorname{rank} D \leq 3$. Moreover, $D = rr^T - P \circ P$ has at most one positive eigenvalue. Since $D \neq 0$, has nonnegative off-diagonal entries, and has trace zero, its nonzero eigenvalues have the form

$$\lambda, \quad -a, \quad -b, \qquad \lambda > 0, \quad a, b \geq 0, \quad a + b = \lambda,$$

where one of a, b may vanish. In particular,

$$\|D\|_{\mathrm{op}} = \lambda, \qquad \|D\|_F^2 = \lambda^2 + a^2 + b^2 \geq \frac{3}{2}\lambda^2.$$

Using $D\mathbf{1} = r$, we obtain

$$\sum_i r_i^2 = \|D\mathbf{1}\|^2 \leq m\lambda^2 \leq \frac{2m}{3}\|D\|_F^2 = \frac{4m}{3}\sum_{i<j} q_{ij}^4.$$

This proves the first inequality in (10). Since

$$\sum_i r_i = 2\sum_{i<j} q_{ij}^2,$$

Cauchy–Schwarz proves the second.

For equality, take

$$x_i = \sqrt{\frac{2}{m}}\cos\frac{\pi i}{m}, \qquad y_i = \sqrt{\frac{2}{m}}\sin\frac{\pi i}{m} \qquad (1 \le i \le m).$$

Then x, y are orthonormal, all $r_i = 2/m$, and the two remaining eigenvalues of D are both $-1/m$. Equality holds in both inequalities. $\square$

Corollary 2.7. *Under the normalization $\sum_{i<j} q_{ij}^2 = 1$, put*

$$p_i = \frac{1}{2}\sum_{j\neq i} q_{ij}^2, \qquad Q = \sum_i p_i^2.$$

Then

$$\sum_{i<j} q_{ij}^4 - \frac{3}{m^2} \ge \frac{3}{m}\left(Q - \frac{1}{m}\right),$$

and consequently

$$\sum_{i<j} q_{ij}^4 - \frac{3}{m^2} \ge \frac{1}{m^2}\min\{mQ-1, 1\}.$$

Proof. The first inequality in (10) is $\sum q_{ij}^4 \ge 3Q/m$, which is the first asserted inequality. Since $mQ \ge 1$,

$$3(mQ-1) \ge \min\{mQ-1, 1\},$$

and the second asserted inequality follows. $\square$

Corollary 2.8. *Let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^TU = I_3$, put $P = UU^T$, and define*

$$d_I = \det P_{I,I}, \qquad s_i = P_{ii}.$$

Then

$$\sum_{|I|=3} d_I^2 \ge \frac{1}{(n-1)^2}\sum_{i=1}^n s_i^2.$$

Proof. Contract the unit decomposable three-form determined by the columns of U with the ith coordinate vector. The result is a decomposable two-form on $\mathbb{R}^{n-1}$, its squares of the entries are the d_I with $i \in I$, and its squared norm is s_i. Theorem 2.6 gives

$$\sum_{I\ni i} d_I^2 \ge \frac{3}{(n-1)^2} s_i^2.$$

Sum over i and observe that each d_I^2 occurs three times. $\square$

Corollary 2.9. *Let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^TU = I_3$, put $P = UU^T$, and define*

$$b_{ij} = \det P_{\{i,j\},\{i,j\}}, \qquad B = \sum_{i<j} b_{ij}^2, \qquad C = \sum_{|I|=3} \det(P_{I,I})^2.$$

Then

$$B \le 2(n-1)C.$$

In particular, $B \le 10C$ when $n = 6$.

Proof. For a fixed i, contract the unit decomposable three-form determined by U with the ith coordinate vector. In the notation of the first inequality in (10), the squared norm of its contraction at j is b_{ij}. Hence

$$\sum_{I \ni i} \det(P_{I,I})^2 \geq \frac{3}{4(n-1)} \sum_{j \neq i} b_{ij}^2.$$

Summing over i counts every term on the left three times and every term on the right twice. Thus

$$3C \geq \frac{3}{2(n-1)} B,$$

which is the asserted identity. $\square$

The preceding inequality has the following quantitative refinement.

Lemma 2.10. *Let $m \geq 3$, and let D be a nonzero real symmetric $m \times m$ matrix with zero trace, rank at most three, and at most one positive eigenvalue. Suppose that its nonzero eigenvalues are*

$$\lambda, -a, -b, \qquad \lambda > 0, \quad a, b \geq 0, \quad a + b = \lambda.$$

Put $r = D\mathbf{1}$ and $\bar{r} = \mathbf{1}^T r/m$. Then

$$\frac{1}{2}\|D\|_F^2 - \frac{3}{4m}\|r\|^2 \geq \frac{1}{4m}\|r - \bar{r}\mathbf{1}\|^2. \tag{11}$$

Proof. Let $\tau = \max\{a, b\}/\lambda \in [1/2, 1]$. The spectral measure of D/λ at the unit vector $\mathbf{1}/\sqrt{m}$ is the law of a real random variable X supported on $[-\tau, 1]$. Put $m_0 = \mathbb{E}X$. Then

$$\|r\|^2 = m\lambda^2 \mathbb{E}X^2, \qquad \|r - \bar{r}\mathbf{1}\|^2 = m\lambda^2 \big(\mathbb{E}X^2 - m_0^2\big).$$

Since $x^2 \leq (1-\tau)x + \tau$ for $-\tau \leq x \leq 1$, we have

$$\begin{aligned} 4\mathbb{E}X^2 - m_0^2 &\leq 4(1-\tau)m_0 + 4\tau - m_0^2 \\ &\leq 4(\tau^2 - \tau + 1). \end{aligned}$$

The second inequality follows by maximizing the quadratic in m_0; its vertex is $m_0 = 2(1-\tau) \in [-\tau, 1]$. Finally,

$$\frac{1}{2}\|D\|_F^2 = \frac{3}{4}\lambda^2 + \frac{1}{4}(a-b)^2,$$

and substitution of the preceding moment inequality is exactly (11). $\square$

Corollary 2.11. *In the notation of Corollary 2.9, put $Q = \sum_i P_{ii}^2$. Then*

$$B \leq \frac{3(n-1)}{2} C + \frac{1}{2(n-1)} Q. \tag{12}$$

In particular, $B \leq 15C/2 + Q/10$ when $n = 6$.

Proof. Put $m = n - 1$. For each i, contract the decomposable three-form determined by U with the ith coordinate vector. Let D_i be the symmetric $m \times m$ matrix whose off-diagonal entries are the squares of the entries of this two-form and whose diagonal is zero. Put

$$C_i = \frac{1}{2}\|D_i\|_F^2, \qquad B_i = \|D_i \mathbf{1}\|^2.$$

The row sums of D_i are b_{ij}, $j \neq i$, and their sum is $2P_{ii}$. If $D_i = 0$, the following inequality is immediate; otherwise Lemma 2.10 gives

$$C_i - \frac{3}{4m} B_i \geq \frac{1}{4m}\left(B_i - \frac{4}{m} P_{ii}^2\right).$$

Summing over i and using

$$\sum_i C_i = 3C, \qquad \sum_i B_i = 2B$$

yields

$$3C - \frac{3}{2m} B \geq \frac{1}{2m} B - \frac{1}{m^2} Q,$$

which is equivalent to (12). $\square$

The next lemma gives the general form of the incidence inequalities. The six-row statement used below is its specialization, but we retain the specialization as a separate proof because its concrete coordinates are needed for the equality analysis.

Lemma 2.12. *Let $n \geq 5$. Let real numbers d_I, indexed by the three-element subsets of $[n]$, satisfy $\sum_I d_I = 1$. Define*

$$p_i = \sum_{I \ni i} d_I, \qquad b_{ij} = \sum_{I \supset \{i,j\}} d_I,$$

and put

$$Q = \sum_i p_i^2, \qquad B = \sum_{i<j} b_{ij}^2, \qquad C = \sum_I d_I^2.$$

Then

$$\frac{4}{n-2} Q - \frac{18}{(n-1)(n-2)} \leq B \leq (n-4)C + \frac{2}{n-3} Q - \frac{6}{(n-2)(n-3)}. \tag{13}$$

More precisely, there are orthogonal subspaces V_2, V_3 of $\mathbb{R}^{\binom{n}{3}}$, depending only on n, such that

$$B - \frac{4}{n-2} Q + \frac{18}{(n-1)(n-2)} = (n-4)\|\Pi_{V_2} d\|_2^2,$$
$$(n-4)C + \frac{2}{n-3} Q - \frac{6}{(n-2)(n-3)} - B = (n-4)\|\Pi_{V_3} d\|_2^2.$$

Proof. Let A and M be the vertex–three-set and pair–three-set incidence matrices, respectively, and let J be the all-one matrix of order $\binom{n}{3}$. Put

$$V_0 = \operatorname{span}\{\mathbf{1}\}, \qquad V_1 = \operatorname{im} A^T \cap V_0^\perp,$$

$$V_2 = \operatorname{im} M^T \cap (V_0 \oplus V_1)^\perp, \qquad V_3 = \ker M.$$

The standard incidence count gives the orthogonal decomposition $\mathbb{R}^{\binom{n}{3}} = V_0 \oplus V_1 \oplus V_2 \oplus V_3$, and the following simultaneous eigenvalue table:

	V_0	V_1	V_2	V_3
$A^T A$	$3\binom{n-1}{2}$	$\binom{n-2}{2}$	0	0
$M^T M$	$3(n-2)$	$2(n-3)$	$n-4$	0
J	$\binom{n}{3}$	0	0	$0.$

Indeed, $AA^T = \binom{n-2}{2} I_n + \binom{n-3}{2} J_n$ and $(M^T M)_{I,K} = \binom{|I \cap K|}{2}$, which yield the table by successively taking orthogonal complements.

The first expression in

$$M^TM - \frac{4}{n-2}A^TA + \frac{18}{(n-1)(n-2)}J,$$

has eigenvalue $n-4$ on V_2 and zero on the other three spaces. The matrix

$$(n-4)I + \frac{2}{n-3}A^TA - \frac{6}{(n-2)(n-3)}J - M^TM$$

has eigenvalue $n-4$ on V_3 and zero on the other three spaces. Taking their quadratic forms at d proves the two displayed identities, and therefore (13). $\square$

Corollary 2.13. *Let $n \geq 5$, let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^TU = I_3$, and put $P = UU^T$. Define*

$$Q = \sum_i P_{ii}^2, \qquad B = \sum_{i<j} \det(P_{\{i,j\},\{i,j\}})^2, \qquad C = \sum_{|I|=3} \det(P_{I,I})^2.$$

Then

$$\frac{4}{n-2}Q - \frac{18}{(n-1)(n-2)} \leq B \leq (n-4)C + \frac{2}{n-3}Q - \frac{6}{(n-2)(n-3)}.$$

Proof. Apply Lemma 2.12 to $d_I = \det(P_{I,I})$. Cauchy–Binet gives $\sum_I d_I = 1$, and the determinantal marginal identities give

$$\sum_{I\ni i} d_I = P_{ii}, \qquad \sum_{I\supset\{i,j\}} d_I = \det(P_{\{i,j\},\{i,j\}}).$$

$\square$

Proposition 2.14. *Let $n > 3$ and let $U \in \mathbb{R}^{n\times 3}$ satisfy $U^TU = I_3$. Then there is a three-element set I such that*

$$\sigma_{\min}(U_I)^2 \geq \frac{2}{3n-4}.$$

The quantities Q, B, C also satisfy a second inequality coming from the incidence relations among vertices, pairs, and three-element sets. We give the linear-algebra argument because this inequality will be used together with the preceding spectral remainder.

Lemma 2.15. *Let nonnegative numbers d_I, indexed by the three-element subsets of $\{1, \dots, 6\}$, satisfy $\sum_I d_I = 1$. Define*

$$p_i = \sum_{I\ni i} d_I, \qquad b_{ij} = \sum_{I\supset\{i,j\}} d_I,$$

and put

$$Q = \sum_i p_i^2, \qquad B = \sum_{i<j} b_{ij}^2, \qquad C = \sum_I d_I^2.$$

Then

$$Q - \frac{9}{10} \leq B \leq 2C + \frac{2}{3}Q - \frac{1}{2}. \tag{14}$$

Proof. Let A be the 6×20 matrix and M the 15×20 matrix defined by

$$A_{i,I} = \mathbf{1}_{\{i\in I\}}, \qquad M_{e,I} = \mathbf{1}_{\{e\subset I\}},$$

where i ranges over the six vertices, e over the fifteen pairs, and I over the twenty three-sets. Let J denote the 20×20 matrix all of whose entries are one. We claim that

$$L := 2I_{20} + \frac{2}{3}A^TA - \frac{1}{2}J - M^TM \succeq 0. \tag{15}$$

this can be checked by the following orthogonal decomposition. Let V_0 be the space of constant vectors, let

$$V_1 = \operatorname{im} A^T \cap V_0^{\perp}, \qquad V_2 = \operatorname{im} M^T \cap (V_0 \oplus V_1)^{\perp}, \qquad V_3 = \ker M.$$

Direct counting gives the dimensions and eigenvalues

	V_0	V_1	V_2	V_3
dim	1	5	9	5
A^TA	30	6	0	0
M^TM	12	6	2	0
J	20	0	0	0.

Indeed, $AA^T = 6I_6 + 4J_6$, while the (I,K) entry of M^TM is $\binom{|I\cap K|}{2}$; applying these two formulas to the four spaces above gives the displayed table. In particular,

$$\mathbb{R}^{20} = V_0 \oplus V_1 \oplus V_2 \oplus V_3,$$

and L has eigenvalue zero on $V_0 \oplus V_1 \oplus V_2$ and eigenvalue two on V_3. This proves (15).

Writing $d = (d_I)_I$, we have

$$Ad = (p_i)_i, \qquad Md = (b_{ij})_{i<j}, \qquad d^TJd = 1.$$

The same table shows that

$$M^TM - A^TA + \frac{9}{10}J$$

has eigenvalue two on V_2 and eigenvalue zero on the other three spaces. Consequently

$$B - Q + \frac{9}{10} = 2\|\Pi_{V_2}d\|^2 \geq 0, \qquad 2C + \frac{2}{3}Q - \frac{1}{2} - B = 2\|\Pi_{V_3}d\|^2 \geq 0,$$

where Π_{V_j} denotes orthogonal projection onto V_j. This proves (14). □

Corollary 2.16. *In the notation of Corollary 2.11, when $n = 6$ one has*

$$Q - \frac{9}{10} \leq B \leq 2C + \frac{2}{3}Q - \frac{1}{2}.$$

Proof. For $d_I = \det P_{I,I}$, Cauchy–Binet gives $\sum_I d_I = 1$, and the determinantal marginal identities give

$$\sum_{I\ni i} d_I = P_{ii}, \qquad \sum_{I\supset\{i,j\}} d_I = \det P_{\{i,j\},\{i,j\}}.$$

Lemma 2.15 applies. □

2.2 Mixed characteristic polynomials

A polynomial with real coefficients is called *real stable* if it does not vanish when all its variables have positive imaginary parts. We use the following multiaffine form of the symbol criterion.

Lemma 2.17. *Let T be a linear operator from multiaffine polynomials in $z_1, \ldots, z_N$ to polynomials in variables x. If*

$$\mathcal{S}_T(x,w) = T\left[\prod_{i=1}^{N}(z_i + w_i)\right]$$

is real stable in $x, w_1, \ldots, w_N$, then T maps every real stable multiaffine polynomial to a real stable polynomial or to zero.

Proof. Polarizing one variable at a time gives. In one variable the assertion is the Hermite–Biehler criterion: the two coefficient polynomials have interlacing real zeros with the proper orientation exactly when their linear symbol is stable. Iteration gives the polarized symbol $\mathcal{S}_T$; diagonalizing the polarized variables gives the stated multiaffine operator. Specialization to real values and passage to limits preserve stability. □

Lemma 2.18. *Let μ be a probability measure on subsets of $[N]$ whose multiaffine generating polynomial*

$$g_\mu(z) = \sum_{S\subseteq[N]} \mu(S) z^S$$

is real stable. Let A_0 be real symmetric, and let $A_i = a_i a_i^T \succeq 0$ have rank one. Then

$$\chi_\mu(x) = \sum_S \mu(S) \det\left(xI - A_0 - \sum_{i\in S} A_i\right)$$

is real-rooted.

Proof. Put

$$F(x,y) = \det\left(xI - A_0 + \sum_{i=1}^N y_i A_i\right) = \sum_{T\subseteq[N]} F_T(x) y^T.$$

This polynomial is multiaffine in y. It is real stable because, when all variables have positive imaginary parts, the imaginary part of the matrix inside the determinant is positive definite. Define

$$T_F h = F(x, -\partial_{z_1}, \dots, -\partial_{z_N}) h(z)|_{z=\mathbf{1}}.$$

The algebraic symbol of this operator is

$$\mathcal{S}_{T_F}(x,w) = \prod_{i=1}^N (1+w_i) F\left(x, -\frac{1}{1+w_1}, \dots, -\frac{1}{1+w_N}\right).$$

If $\operatorname{Im} w_i > 0$, then $\operatorname{Im}[-(1+w_i)^{-1}] > 0$. Hence the preceding symbol is real stable, and Lemma 2.17 shows that $T_F g_\mu$ is real stable or zero.

For every $T \subseteq [N]$,

$$\partial^T g_\mu(\mathbf{1}) = \Pr_\mu(T \subseteq S).$$

Expanding first in T and then in S gives

$$T_F g_\mu = \sum_T (-1)^{|T|} F_T(x) \Pr_\mu(T\subseteq S) = \chi_\mu(x).$$

The polynomial is monic, so it is not zero. As a stable univariate polynomial with real coefficients, it is real-rooted. □

Proof of Proposition 2.14. Write u_i^T for the ith row of U, put $P = UU^T$, and define

$$d_I = \det P_{I,I} \qquad (|I| = 3).$$

The numbers d_I form a probability distribution, and its generating polynomial is

$$g(z) = \sum_{|I|=3} d_I z^I = \det\left(U^T \operatorname{diag}(z_1,\dots,z_n) U\right).$$

It is real stable: when all z_i have positive imaginary parts, the imaginary part of the matrix in the preceding determinant is positive definite.

Take $A_i = u_i u_i^T$. We first show that the mixed characteristic polynomials form a interlacing family. Order the coordinates and condition successively on their inclusion indicators. Setting a variable of g to zero conditions on exclusion; differentiating in that variable and then setting it to zero conditions on inclusion. Both operations preserve real stability. For the two children at any node, every nonnegative linear combination of their unnormalized mixed characteristic polynomials is obtained by multiplying the weights of sets containing the next coordinate by a nonnegative multiplier. This is a nonnegative rescaling of one variable of the conditional generating polynomial, and is real stable by Lemma 2.18. Hence the two child polynomials have a common interlacing. Recursively choosing a child whose smallest zero is no smaller than that of its parent yields a three-set I such that

$$\lambda_{\min}(P_{I,I}) \geq r_1, \tag{16}$$

where r_1 is the smallest zero of the resulting polynomial.

Put

$$p_i = P_{ii}, \qquad b_{ij} = \det P_{\{i,j\},\{i,j\}},$$

$$Q = \sum_i p_i^2, \qquad B = \sum_{i<j} b_{ij}^2, \qquad C = \sum_{|I|=3} d_I^2.$$

The determinantal inclusion identities give

$$\Pr(i \in I) = p_i, \qquad \Pr(\{i,j\} \subseteq I) = b_{ij}.$$

Consequently the polynomial is

$$\chi(x) = x^3 - Qx^2 + Bx - C. \tag{17}$$

Let its zeros be $r_1 \leq r_2 \leq r_3$. They lie in $(0, 1]$. Indeed, real-rootedness and the alternating positive coefficients in (17) exclude negative zeros. Moreover,

$$-\chi(1-y) = \sum_I d_I \det\left(yI_3 - \left(I_3 - U_I^T U_I\right)\right)$$

has alternating nonnegative coefficients because $0 \preceq U_I^T U_I \preceq I_3$; its zeros $1 - r_j$ are therefore nonnegative.

By Vieta's formulas and Corollary 2.9,

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{B}{C} \leq 2(n-1).$$

Since $r_2, r_3 \leq 1$,

$$\frac{1}{r_1} \leq 2(n-1) - 2 = 2n - 4.$$

Thus $r_1 \geq 1/(2n-4)$. Put

$$m = n - 1, \qquad x = r_1, \qquad y = r_2, \qquad z = r_3,$$

and suppose, for a contradiction, that

$$x < \frac{2}{3m-1}.$$

The case $m = 3$ is already excluded by the preceding lower bound, so assume $m \geq 4$. Since $Q = \sum_i P_{ii}^2$ and $\sum_i P_{ii} = 3$,

$$Q = x + y + z \geq \frac{9}{m+1}.$$

Corollary 2.11 gives

$$F(x,y,z) := xy + xz + yz - \frac{3m}{2}xyz - \frac{x+y+z}{2m} \le 0. \tag{18}$$

Set $s = y+z$, $t = yz$,

$$A = x - \frac{1}{2m} > 0, \qquad c = 1 - \frac{3m}{2}x.$$

Then

$$F = As - \frac{x}{2m} + ct.$$

Suppose first that $c \ge 0$, equivalently $x \le 2/(3m)$. Since $(y-x)(z-x) \ge 0$, one has $t \ge x(s-x)$. Both coefficients of s in the resulting lower bound are positive, while $s \ge 9/(m+1) - x$. Hence

$$F \ge \frac{3(mx-1)\big(2m^2x^2 + 2mx^2 - 9mx + 3\big)}{2m(m+1)}.$$

Here

$$\frac{m}{2m-2} \le mx \le \frac{2}{3}.$$

Thus $mx - 1 < 0$. The second factor in parentheses is also negative: as a function of $u = mx$ it equals $2(1+1/m)u^2 - 9u + 3$, which is decreasing on this interval and is already negative at $u = 1/2$. Therefore the preceding expression is positive, contrary to (18).

Consider $c < 0$. Since $t \le s^2/4$,

$$F \ge K(s) := As - \frac{x}{2m} + \frac{c}{4}s^2.$$

The function K is concave, and

$$\frac{9}{m+1} - x \le s \le 2.$$

It is therefore enough to check the two endpoints. At $s = 2$,

$$K(2) = \frac{(m-1)(2-(3m-1)x)}{2m} > 0.$$

At the other endpoint, direct factorization gives

$$\begin{aligned} K\left(\frac{9}{m+1} - x\right) &= -\frac{3}{8m(m+1)^2}\big(m(m+1)x - 7m + 2\big) \\ &\quad \cdot \big(m(m+1)x^2 - 9mx + 6\big). \end{aligned} \tag{19}$$

On $2/(3m) < x < 2/(3m-1)$, the first parenthesis is negative. The second is decreasing in x and, at the right endpoint, equals

$$\frac{2(m-3)(2m-1)}{(3m-1)^2} > 0.$$

Thus the preceding factorization is positive as well, again a contradiction. We conclude that $r_1 \ge 2/(3n-4)$. Combining this with (16) and (5) proves the theorem. $\square$

We shall also use the real $k = 2$ case of (1).

Theorem 2.19 (Sengupta–Pautov [5]). *If $V \in \mathbb{R}^{N\times 2}$ and $V^TV = I_2$, then two rows of V form a 2×2 matrix R satisfying $\sigma_{\min}(R) \ge N^{-1/2}$.*

3 General Lower Bounds for Arbitrary n

Proof of Theorem 1.1. Put

$$D_n = (n-2)(n^2+n-4), \qquad A_n = 5n^2 - 11n + 4,$$

and

$$h_n(t) = D_n t^2 - A_n t + 3n - 4.$$

The discriminant of h_n is

$$\Delta_n = (n^2-3n+4)(13n^2-43n+36),$$

so τ_n is its smaller zero. In the notation of Corollaries 2.11 and 2.13, define

$$\begin{aligned}
\mathcal{U} &= B - (n-4)C - \frac{2}{n-3}Q + \frac{6}{(n-2)(n-3)}, \\
\mathcal{S} &= B - \frac{3(n-1)}{2}C - \frac{1}{2(n-1)}Q, \\
\mathcal{R} &= C + Q - B - 1.
\end{aligned}$$

The two preceding inequalities give $\mathcal{U} \le 0$ and $\mathcal{S} \le 0$. If $r_1 \le r_2 \le r_3$ are the zeros of the mixed polynomial (17), then $0 < r_j \le 1$, and hence

$$-\mathcal{R} = 1 - Q + B - C = \prod_{j=1}^{3}(1-r_j) \ge 0.$$

For a variable t, put

$$\begin{aligned}
\lambda_U(t) &= -\frac{(n-3)(t-1)\big((3n^2-8n+5)t-2n+3\big)}{(n-5)(n^2+n-4)}, \\
\lambda_S(t) &= \frac{2(n-1)(t-1)((n-3)t-1)}{n^2+n-4}, \\
\lambda_R(t) &= -\frac{N_n(t)}{(n-5)(n^2+n-4)},
\end{aligned}$$

where

$$N_n(t) = (n^3+n^2-17n+15)t^2 + (-5n^2+5n+6)t + 3n - 1.$$

Matching the coefficients of Q, B, C gives the exact identity

$$\begin{aligned}
t^3 - Qt^2 + Bt - C = \lambda_U(t)\mathcal{U} + \lambda_S(t)\mathcal{S} + \lambda_R(t)\mathcal{R} \qquad (20) \\
+ \frac{(t-1)h_n(t)}{(n-2)(n^2+n-4)}. \qquad (21)
\end{aligned}$$

We check the signs at $t = \tau_n$. Direct calculation gives

$$h_n\left(\frac{2n-3}{3n^2-8n+5}\right) = \frac{(n-2)^3(n^2+n-4)}{(n-1)^2(3n-5)^2} > 0,$$

$$h_n\left(\frac{1}{n-3}\right) = -\frac{(n-4)(n^2+n-4)}{(n-3)^2} < 0.$$

It follows that

$$\frac{2n-3}{3n^2-8n+5} < \tau_n < \frac{1}{n-3},$$

and therefore $\lambda_U(\tau_n), \lambda_S(\tau_n) \geq 0$. To check the remaining multiplier, substitution gives

$$N_n(\tau_n) = \frac{P_n - Q_n\sqrt{\Delta_n}}{2(n-2)^2(n^2+n-4)^2},$$

where

$$P_n = 14n^6 - 267n^5 + 1431n^4 - 3713n^3 + 5319n^2 - 4056n + 1264,$$
$$Q_n = 4n^4 - 61n^3 + 190n^2 - 229n + 108.$$

The comparison needed here follows from

$$Q_n^2\Delta_n - P_n^2 = 4(n-2)^2(n^2+n-4)^3(3n^4 - 42n^3 + 75n^2 - 8n - 80).$$

For $6 \leq n \leq 11$, one has $P_n, Q_n < 0$ and the last factor in the preceding factorization is negative; for $n = 12$, $P_n < 0 < Q_n$; and for $n \geq 13$, all three quantities P_n, Q_n and the last factor are positive. The last assertion follows immediately after expanding at $n = 13$. In all cases

$$P_n - Q_n\sqrt{\Delta_n} < 0.$$

Thus $N_n(\tau_n) < 0$ and $\lambda_R(\tau_n) > 0$.

Since $h_n(\tau_n) = 0$, the dual certificate (21) now yields

$$\chi(\tau_n) \leq 0.$$

Also

$$h_n\left(\frac{1}{n-1}\right) = -\frac{(n-2)(n^2-3n+4)}{(n-1)^2} < 0,$$

so $\tau_n < 1/(n-1)$. Corollary 2.9 gives

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{B}{C} \leq 2(n-1).$$

If $r_1 < \tau_n$ and $r_2 \leq \tau_n$, the first two reciprocal terms already exceed $2/\tau_n > 2(n-1)$, which is impossible. Hence $r_1 < \tau_n$ would imply $r_1 < \tau_n < r_2$ and therefore $\chi(\tau_n) > 0$, contradicting the preceding bound. Thus $r_1 \geq \tau_n$, and the mixed interlacing selection proves $\Lambda_n \geq \tau_n$.

Finally,

$$h_n\left(\frac{2}{3n-4}\right) = \frac{n(n-4)(n-2)}{(3n-4)^2} > 0,$$

whereas $h_n(1/(n-3)) < 0$. Since $2/(3n-4) < 1/(n-3)$, the smaller zero satisfies $\tau_n > 2/(3n-4)$. □

Lemma 3.1. *Let $2/3 \leq x \leq y \leq z$. Put*

$$s = x + y + z, \qquad q = xy + xz + yz, \qquad p = xyz.$$

If

$$q \leq \frac{3}{2}p + \frac{1}{2}s, \qquad q \leq p + 2s - 6, \tag{22}$$

then

$$x \geq \frac{5 - \sqrt{13}}{2}. \tag{23}$$

Proof. Write
$$\alpha = \frac{5-\sqrt{13}}{2}.$$
Suppose first that $x = 2/3$. The two inequalities in (22) become
$$z \le 2-y, \qquad z \ge \frac{14-4y}{4-y}.$$
The first inequality and $z \ge y$ give $y \le 1$, whereas on this interval the second lower bound is greater than one. This is impossible.

It remains to exclude $2/3 < x < \alpha$. Define
$$L_1 = x+y-\frac{3}{2}xy-\frac{1}{2}, \qquad N_1 = \frac{x+y}{2}-xy,$$
$$L_2 = x+y-xy-2, \qquad N_2 = 2(x+y)-6-xy.$$
The two assumptions are exactly
$$zL_1 \le N_1, \qquad zL_2 \le N_2. \tag{24}$$
Put
$$r_1 = \frac{x-1/2}{3x/2-1}, \qquad r_2 = \frac{2-x}{1-x}, \qquad y_0 = \frac{x}{2x-1}.$$
Since $x^2-5x+3>0$ and $x^2-4x+2<0$ on the interval under consideration, direct subtraction gives
$$r_1 > r_2 > y_0. \tag{25}$$
Moreover, $L_1 > 0$ exactly when $y < r_1$, and $L_2 > 0$ exactly when $y > r_2$.

For later use set
$$D(x,y) = N_1L_2 - N_2L_1.$$
As a polynomial in y, it is concave, with leading coefficient $-(x^2-5x+3)/2$. At the three relevant points one has
$$D(x,x) = -\frac{x-1}{2}\left(x^3-9x^2+18x-6\right) > 0, \tag{26}$$
$$D(x,y_0) = -\frac{3(x-1)^2(x^2-4x+2)}{2(2x-1)^2} > 0, \tag{27}$$
$$D(x,r_1) = -\frac{(x-1)^2(x^2-5x+3)}{2(3x-2)^2} < 0.$$
The first sign follows, for example, because the cubic in (26) is positive and increasing on $[2/3,\alpha]$. Furthermore,
$$\partial_y D(x,r_1) = \frac{11x^3-51x^2+56x-18}{2(3x-2)} < 0.$$
Indeed, the numerator is increasing on $[2/3,\alpha]$ and its value at α is $43\alpha-30<0$. Concavity, the displayed value of $D(x,r_1)$, and the derivative estimate above therefore show that
$$D(x,y) < 0 \qquad (y \ge r_1). \tag{28}$$

If $y = r_1$, then $L_1 = 0$ while
$$N_1 = -\frac{(x-1)^2}{2(3x-2)} < 0,$$
contradicting the first inequality in (24). If $y > r_1$, then $L_1 < 0 < L_2$; the existence of a common z satisfying (24) would require $D(x,y) \ge 0$, contrary to (28).

Finally suppose that $y < r_1$. Then $L_1 > 0$. If $y > y_0$, one has $N_1 < 0$, which again makes the first inequality in (24) impossible. Hence $x \le y \le y_0$. By (25), $L_2 < 0$. A common z would now require $D(x, y) \le 0$. But concavity together with (26) and (27) gives $D(x, y) > 0$ throughout $[x, y_0]$. This final contradiction proves (23). $\square$

Proof of Theorem 1.2. Suppose the conclusion is false. By compactness of the Stiefel manifold, there are integers $n \to \infty$ and three-frames U_n for which

$$n \max_{|I|=3} \sigma_{\min}((U_n)_I)^2 \le \alpha - \varepsilon, \qquad \alpha = \frac{5 - \sqrt{13}}{2},$$

for some $\varepsilon > 0$. Let $0 < x_n \le y_n \le z_n \le 1$ be the roots of the polynomial χ_n (17). The interlacing selection (16) gives $nx_n \le \alpha - \varepsilon$, whereas Proposition 2.14 gives

$$nx_n \ge \frac{2n}{3n - 4}.$$

Put

$$X_n = nx_n, \qquad Y_n = ny_n, \qquad Z_n = nz_n,$$

and

$$S_n = X_n + Y_n + Z_n, \qquad T_n = X_nY_n + X_nZ_n + Y_nZ_n, \qquad R_n = X_nY_nZ_n.$$

After passage to a subsequence, $X_n \to X$ with

$$\frac{2}{3} \le X < \alpha. \tag{29}$$

Corollary 2.11 and Corollary 2.13, after multiplication by n^2, give

$$T_n \le \frac{3}{2}\left(1 - \frac{1}{n}\right) R_n + \frac{n}{2(n-1)} S_n, \tag{30}$$

$$T_n \le \left(1 - \frac{4}{n}\right) R_n + \frac{2n}{n-3} S_n - \frac{6n^2}{(n-2)(n-3)}. \tag{31}$$

If (Z_n) is bounded, pass to a further subsequence on which $Y_n \to Y$ and $Z_n \to Z$. Equations (30)–(31) then give

$$XY + XZ + YZ \le \frac{3}{2}XYZ + \frac{1}{2}(X + Y + Z),$$

$$XY + XZ + YZ \le XYZ + 2(X + Y + Z) - 6.$$

Since $X \le Y \le Z$, Lemma 3.1 contradicts (29).

Consider $Z_n \to \infty$. If also $Y_n \to \infty$, divide (31) by Y_nZ_n. Its left side tends to one, while its right side tends to $X < 1$, a contradiction. Hence (Y_n) is bounded; pass to a subsequence with $Y_n \to Y$. Dividing (30) and (31) by Z_n gives, respectively,

$$X + Y \le \frac{3}{2}XY + \frac{1}{2}, \qquad X + Y \le XY + 2.$$

The first inequality excludes $X = 2/3$. For $X > 2/3$, they imply

$$Y \ge \frac{X - 1/2}{3X/2 - 1}, \qquad Y \le \frac{2 - X}{1 - X}.$$

Their compatibility is equivalent to $X^2 - 5X + 3 \le 0$, and hence $X \ge \alpha$, again contradicting (29). This proves (2). $\square$

4 The Five-Line Restriction

We begin with a weighted consequence of Theorem 2.19.

Lemma 4.1. *Let $y_1, \ldots, y_m \in \mathbb{R}^2$ and let $p_1, \ldots, p_m > 0$. Suppose*

$$B = \sum_{i=1}^{m} p_i y_i y_i^T \succ 0, \qquad \sum_{i=1}^{m} p_i \le 1.$$

Then there are distinct indices j, k such that

$$y_j y_j^T + y_k y_k^T \succeq B.$$

Proof. Replacing y_i by $B^{-1/2} y_i$, it is enough to prove the statement when $B = I_2$.

Suppose first that all p_i are rational. Choose $N \in \mathbb{N}$ and nonnegative integers N_i such that $p_i = N_i/N$. Form an $N \times 2$ matrix by taking N_i copies of $y_i^T/\sqrt{N}$ and then adding $N - \sum_i N_i$ zero rows. Its columns are orthonormal. By Theorem 2.19, two rows form a matrix with smallest singular value at least $N^{-1/2}$. Neither row is zero, and the two rows cannot be copies associated with the same index, since such a matrix would be singular. Thus they correspond to distinct j, k, and

$$y_j y_j^T + y_k y_k^T \succeq I_2.$$

For arbitrary positive weights, choose positive rational vectors $p^{(r)} \to p$ such that $\sum_i p_i^{(r)} \le 1$. Put

$$B_r = \sum_i p_i^{(r)} y_i y_i^T, \qquad y_i^{(r)} = B_r^{-1/2} y_i.$$

For all sufficiently large r, $B_r \succ 0$. Apply the rational case to $y_i^{(r)}$ and $p_i^{(r)}$. Since there are finitely many pairs, one pair occurs along a subsequence. Letting $r \to \infty$ proves the result. $\square$

We first treat three or four supporting lines directly.

Proposition 4.2. *The conclusion of Theorem 1.3 holds if the nonzero rows of U are contained in the union of at most four lines.*

Proof. Since U has rank three, at least three lines are required. Partition the nonzero rows according to their supporting lines. In group j, write the rows as

$$\alpha_{j,\ell} v_j, \qquad \ell = 1, \ldots, m_j,$$

where $\|v_j\| = 1$. Define

$$s_j = \sum_{\ell=1}^{m_j} \alpha_{j,\ell}^2, \qquad w_j = \sqrt{s_j}\, v_j, \qquad a_j = \max_{\ell} \alpha_{j,\ell}^2, \qquad c_j = \frac{a_j}{s_j}.$$

If W is the matrix with rows w_j^T, then $W^T W = I_3$, and

$$c_j \ge \frac{1}{m_j}, \qquad \sum_j \frac{1}{c_j} \le \sum_j m_j = n.$$

If there are three groups, W is orthogonal. Selecting a row of squared norm a_j from every group gives, up to row signs,

$$Q = \operatorname{diag}(\sqrt{c_1}, \sqrt{c_2}, \sqrt{c_3}) W.$$

Because $c_j \ge 1/m_j \ge 1/n$, we have $QQ^T \succeq I_3/n$.

Suppose there are four groups. Complete W to an orthogonal matrix $[W\ z] \in \mathbb{R}^{4\times 4}$, so

$$WW^T = I_4 - zz^T.$$

Put $t = 1/n$, $x_j = z_j^2$, and $b_j = t/(c_j - t)$. Since $m_j < n$, we have $c_j > t$. Set

$$y_j = (1 + b_j)x_j, \qquad B_0 = \sum_j b_j x_j.$$

Using $b_j/(1 + b_j) = t/c_j$ and the bounds established above,

$$B_0 \le \left(\max_j y_j\right) \sum_j \frac{t}{c_j} \le \max_j y_j.$$

Choose q such that $y_q = \max_j y_j$. Then

$$x_q \ge \sum_{j\neq q} b_j x_j.$$

Select a row of squared norm a_j from each group $j \neq q$. With $D_q = \operatorname{diag}(\sqrt{c_j} : j \neq q)$, its Gram matrix is

$$QQ^T = D_q(I_3 - z_{-q} z_{-q}^T)D_q.$$

Lemma 2.4 and the preceding inequality give

$$\sum_{j\neq q} \frac{c_j x_j}{c_j - t} = 1 - x_q + \sum_{j\neq q} b_j x_j \le 1.$$

Hence $QQ^T \succeq tI_3$. $\square$

Proof of Theorem 1.3. By Proposition 4.2, we consider exactly five nonzero groups. Define m_i, s_i, w_i, a_i, c_i as in the proof of that proposition. Let $W \in \mathbb{R}^{5\times 3}$ have rows w_i^T. Then $W^T W = I_3$, and

$$c_i > \frac{1}{n}, \qquad \sum_{i=1}^{5} \frac{1}{c_i} \le n.$$

Complete W to an orthogonal matrix $[W\ Z] \in \mathbb{R}^{5\times 5}$, where $Z \in \mathbb{R}^{5\times 2}$, and denote the ith row of Z by z_i^T. Thus

$$WW^T = I_5 - ZZ^T, \qquad \sum_{i=1}^{5} z_i z_i^T = I_2.$$

Set $t = 1/n$ and

$$b_i = \frac{t}{c_i - t}, \qquad y_i = \sqrt{1 + b_i}\, z_i, \qquad p_i = \frac{t}{c_i}.$$

By the preceding bounds, $\sum_i p_i \le 1$, and

$$B := \sum_i p_i y_i y_i^T = \sum_i b_i z_i z_i^T.$$

If $B \succ 0$, Lemma 4.1 gives a pair $J = \{j, k\}$ such that

$$\sum_{i\in J} (1 + b_i) z_i z_i^T \succeq B. \tag{32}$$

If B is singular, all z_i belong to its one-dimensional range, and (32) follows from the corresponding scalar assertion: if $B = \sum_i p_i \eta_i^2$ and $\sum_i p_i \le 1$, then some j satisfies $\eta_j^2 \ge B$, after which any distinct k may be added. Thus (32) holds in both cases.

Let $I = \{1, \dots, 5\} \setminus J$. Select a row of squared norm a_i from each group indexed by I. Up to row signs, the Gram matrix of the resulting 3×3 matrix Q is

$$QQ^T = D_I(I_3 - Z_I Z_I^T)D_I, \qquad D_I = \operatorname{diag}(\sqrt{c_i} : i \in I).$$

By Lemma 2.4, $QQ^T \succeq tI_3$ is equivalent to

$$\sum_{i \in I} \frac{c_i}{c_i - t} z_i z_i^T \preceq I_2.$$

Using the preceding bounds and (32), the left-hand side is

$$\sum_{i \in I} (1 + b_i) z_i z_i^T = I_2 + B - \sum_{i \in J} (1 + b_i) z_i z_i^T \preceq I_2.$$

Therefore $QQ^T \succeq I_3/n$, which is equivalent to the asserted bound. □

Corollary 4.3. *If at least $n - 3$ rows of $U \in \mathbb{R}^{n \times 3}$ lie on one line, then the conclusion of* (1) *holds.*

Proof. The remaining three rows add at most three further supporting lines, so Proposition 4.2 applies. □

Proposition 4.4. *For every $n \geq 4$, there is a matrix $U \in \mathbb{R}^{n \times 3}$ with $U^T U = I_3$ such that*

$$\max_{|I|=3} \sigma_{\min}(U_I)^2 = \frac{1}{n}.$$

The nonzero rows of U lie on four lines.

Proof. Choose $u_1, u_2, u_3 \in \mathbb{R}^2$ satisfying

$$\|u_j\|^2 = \frac{2}{3}, \qquad \langle u_i, u_j \rangle = -\frac{1}{3} \quad (i \neq j), \qquad \sum_{j=1}^{3} u_j = 0.$$

For example, one may take

$$u_1 = \sqrt{\frac{2}{3}}(1, 0), \quad u_2 = \sqrt{\frac{2}{3}}\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad u_3 = \sqrt{\frac{2}{3}}\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Put

$$a^2 = \frac{n-1}{n(n-3)}, \qquad b^2 = \frac{1}{3n}.$$

Take $n - 3$ rows equal to $x = (a, 0, 0)$ and the remaining rows equal to

$$y_j = (b, u_j), \qquad j = 1, 2, 3.$$

The displayed formula gives

$$(n-3)xx^T + \sum_{j=1}^{3} y_j y_j^T = I_3,$$

so $U^T U = I_3$.

The triple (y_1, y_2, y_3) has covariance matrix

$$\sum_{j=1}^{3} y_j y_j^T = \operatorname{diag}(1/n, 1, 1),$$

and therefore has squared smallest singular value $1/n$.

For $i \neq j$, the Gram matrix of (x, y_i, y_j) has eigenvalue 1 in the antisymmetric direction of the last two coordinates. On the orthogonal two-dimensional invariant subspace it is represented by

$$\begin{pmatrix} a^2 & \sqrt{2}ab \\ \sqrt{2}ab & 2b^2 + 1/3 \end{pmatrix}.$$

At $t = 1/n$,

$$(a^2 - t)(2b^2 + 1/3 - t) - 2a^2b^2 = 0,$$

while its trace is greater than $2t$. Its smaller eigenvalue is therefore t. Every triple containing at least two copies of x is singular. This proves the stated maximum. The rows x, y_1, y_2, y_3 determine four distinct lines. $\square$

This proves Theorem 1.4.

Proof of Theorem 1.5. The case $k = 1$ is immediate. Suppose $k \geq 2$. Let $u_1, \ldots, u_k \in \mathbb{R}^{k-1}$ be the vertices of a regular simplex centered at the origin, normalized by

$$\sum_{j=1}^{k} u_j u_j^T = I_{k-1}, \qquad \langle u_i, u_j \rangle = -\frac{1}{k} \quad (i \neq j).$$

Thus $\|u_j\|^2 = (k-1)/k$ and $\sum_j u_j = 0$. Set

$$a^2 = \frac{n-1}{n(n-k)}, \qquad b^2 = \frac{1}{kn}.$$

Take $n - k$ rows equal to $x = (a, 0, \ldots, 0)$ and k further rows

$$y_j = (b, u_j) \qquad (1 \leq j \leq k).$$

Their outer products sum to I_k, so the resulting matrix V has orthonormal columns.

Every nonsingular k-row submatrix consists either of all the y_j or of one copy of x and $k-1$ of the y_j. The first type has column Gram matrix $\operatorname{diag}(1/n, 1, \ldots, 1)$. For the second type, omit y_q and restrict the column Gram matrix to the span of the first coordinate vector and u_q. In the corresponding orthonormal basis it is

$$\begin{pmatrix} a^2 + (k-1)b^2 & -b\sqrt{(k-1)/k} \\ -b\sqrt{(k-1)/k} & 1/k \end{pmatrix}.$$

Subtracting I_2/n gives determinant zero, because

$$\left(a^2 + (k-1)b^2 - \frac{1}{n}\right)\left(\frac{1}{k} - \frac{1}{n}\right) = \frac{k-1}{k^2 n} = b^2 \frac{k-1}{k}.$$

The remaining $k-2$ eigenvalues are equal to 1, and the other eigenvalue of the displayed two-by-two matrix exceeds $1/n$. Hence both nonsingular types have smallest squared singular value $1/n$. Any selected set containing two copies of x is singular. This proves (3). $\square$

5 Possible Counterexample Scenarios

Proposition 5.1. *Let $n > 3$, let $U \in \mathbb{R}^{n \times 3}$ satisfy $U^T U = I_3$, and put $P = UU^T$. If*

$$\max_{1 \leq i \leq n} P_{ii} \geq 1 - \frac{1}{n} + \frac{1}{n^2}, \tag{33}$$

then there is a three-element set I such that

$$P_{I,I} \succeq \frac{1}{n} I_3.$$

Lemma 5.2. *Let $(v_i)_{i\in E}$ be a finite family of vectors, and let $r(A)$ denote the dimension of the span of $(v_i)_{i\in A}$. The largest cardinality of a subset of E which is the disjoint union of two linearly independent subsets is*

$$\min_{A\subseteq E}\bigl(|E\setminus A|+2r(A)\bigr).$$

Proof. If I_1, I_2 are disjoint and linearly independent, then for every $A\subseteq E$,

$$|I_1\cup I_2|\le |E\setminus A|+|I_1\cap A|+|I_2\cap A|\le |E\setminus A|+2r(A).$$

This proves one inequality. The reverse inequality is the rank formula for the union of two copies of the vector matroid on E, equivalently the two-family case of the matroid union theorem [2]. Applying that formula at the full ground set E gives exactly the asserted formula. ☐

Proposition 5.3. *Let $U\in\mathbb{R}^{n\times 3}$ satisfy $U^TU=I_3$, and put*

$$d_I=\det(U_IU_I^T)\qquad (|I|=3).$$

If

$$d_Id_J=0\qquad \textit{whenever } I\cap J=\varnothing,$$

then there is a three-element set I such that

$$U_IU_I^T\succeq\frac{1}{n}I_3.$$

Proof. Apply Lemma 5.2 to the row vectors of U. The displayed condition says that there are no two disjoint linearly independent triples. Hence there is a set $A\subseteq[n]$ such that

$$|[n]\setminus A|+2r(A)\le 5. \tag{34}$$

If $r(A)=0$, all nonzero rows occur among at most five indices. If $r(A)=1$, all but at most three rows lie on one line. In both cases all nonzero rows are contained in the union of at most five lines, and Theorem 1.3 applies.

If $r(A)=2$, equation (34) shows that all but at most one row lie in a plane L. Since U has rank three, there is exactly one row outside L. Decompose $\mathbb{R}^3=L\oplus L^\perp$. The off-diagonal block of $U^TU=I_3$ shows that the exceptional row is orthogonal to L, and the $L^\perp$ diagonal entry shows that it has norm one. The remaining rows form a Parseval frame in L. The real two-column theorem supplies two of them whose two-dimensional Gram matrix is at least $I_2/(n-1)$. Together with the exceptional unit row they form a three-row submatrix whose Gram matrix is at least I_3/n.

The case $r(A)=3$ is incompatible with (34). This completes the proof. ☐

Proof of Proposition 5.1. Choose an index attaining the maximum in (33). After permuting the rows and applying an orthogonal change of coordinates in $\mathbb{R}^3$, write

$$u_1=(\sqrt{t},0,0),\qquad u_i=(a_i,v_i),\quad v_i\in\mathbb{R}^2\quad (2\le i\le n),$$

where $t=P_{11}$. The Parseval identities give

$$\sum_{i=2}^n a_i^2=1-t,\qquad \sum_{i=2}^n v_iv_i^T=I_2,\qquad \sum_{i=2}^n a_iv_i=0.$$

The real two-column case of the conjecture, applied to the $(n-1)\times 2$ Parseval frame with rows v_i, gives two indices j,k such that

$$C:=v_jv_j^T+v_kv_k^T\succeq\frac{1}{n-1}I_2.$$

Put

$$s = a_j^2 + a_k^2, \qquad b = a_j v_j + a_k v_k.$$

Then $s \le 1 - t$ and the matrix Cauchy–Schwarz inequality gives $bb^T \preceq sC$. The column Gram matrix of the three selected rows is

$$G = \begin{pmatrix} t+s & b^T \\ b & C \end{pmatrix}.$$

Set $r = 1/n$. Since $t + s - r > 0$, its Schur complement at r satisfies

$$\begin{aligned} C - rI_2 - \frac{bb^T}{t+s-r} &\succeq \frac{t-r}{t+s-r} C - rI_2 \\ &\succeq \frac{t-r}{1-r} \frac{1}{n-1} I_2 - rI_2 \\ &= \left(\frac{nt-1}{(n-1)^2} - \frac{1}{n} \right) I_2. \end{aligned}$$

The last coefficient is nonnegative exactly when

$$t \ge \frac{n^2 - n + 1}{n^2} = 1 - \frac{1}{n} + \frac{1}{n^2}.$$

Thus $G \succeq I_3/n$. The nonzero eigenvalues of G agree with those of the principal row Gram matrix $P_{\{1,j,k\},\{1,j,k\}}$, proving the claim. □

We first treat projections whose diagonal is sufficiently far from the constant vector $(1/2, \dots, 1/2)$. The argument uses the ten complementary pairs of three-element subsets of $\{1, \dots, 6\}$.

Proposition 5.4. *Let $P \in \mathbb{R}^{6\times 6}$ be an orthogonal projection of rank three. If*

$$\max_i \left| P_{ii} - \frac{1}{2} \right| \ge \frac{13}{36},$$

then there is a three-element set I such that $P_{I,I} \succeq I_3/6$.

Proof. When $\max_i P_{ii} \ge 31/36$, the assertion is Proposition 5.1 with $n = 6$, since

$$1 - \frac{1}{6} + \frac{1}{36} = \frac{31}{36}.$$

If instead some $P_{ii} \le 5/36$, apply the result just proved to the rank-three projection $I_6 - P$, whose corresponding diagonal entry is at least $31/36$. For complementary three-sets J, J^c, the spectra of $(I_6 - P)_{J,J}$ and P_{J^c,J^c} agree; this follows, for example, from the cosine–sine decomposition of the two complementary coordinate compressions. Hence a good compression for $I_6 - P$ gives a good compression for P. The two alternatives are exactly $|P_{ii} - 1/2| \ge 13/36$. □

Lemma 5.5. *Let G be a simple graph on five vertices, with $m \le 5$ edges and degrees $e_1, \dots, e_5$. Define*

$$c_6 = 5 - m, \qquad c_i = m - 1 - 2e_i \quad (1 \le i \le 5).$$

Then

$$\sum_{i=1}^{6} c_i^2 \le \begin{cases} 30, & m = 0, \\ 24, & m = 1, \\ 22, & m = 2, \\ 24, & m = 3, \\ 30, & m = 4, \\ 24, & m = 5. \end{cases}$$

Equality with value 30 occurs only for $m = 0$, or for $m = 4$ when G is a star with four edges.

Proof. Since $\sum_i e_i = 2m$, direct expansion gives

$$\sum_{i=1}^{6} c_i^2 = (5-m)^2 + 5(m-1)^2 - 8m(m-1) + 4\sum_{i=1}^{5} e_i^2.$$

For $m = 0, 1$, the degree sequences are respectively $(0,0,0,0,0)$ and $(1,1,0,0,0)$. For $m = 2$, two edges are either adjacent or disjoint, so $\sum_i e_i^2 \le 6$. For $m = 3$, either the maximum degree is at most two, in which case $\sum_i e_i^2 \le 2\sum_i e_i = 12$, or the maximum degree is three, which gives the same bound.

Suppose $m = 4$. If the maximum degree is four, the graph is the four-edge star and $\sum_i e_i^2 = 20$. If the maximum degree is three, delete a vertex of degree three. The remaining graph has one edge; its possible position gives $\sum_i e_i^2 \le 18$. If every degree is at most two, then $\sum_i e_i^2 \le 16$. Thus the maximum is 20, attained only by the star. Finally, suppose $m = 5$. If the maximum degree is four, deleting such a vertex leaves one edge and the degree sequence is $(4,2,2,1,1)$, whose squared sum is 26. If the maximum degree is three, deletion leaves two edges and $\sum_i e_i^2 \le 24$; if every degree is at most two, the squared sum is at most 20. Substitution of these six bounds in the preceding degree identity proves the stated values and the equality assertion. $\square$

Proposition 5.6. *Let $P \in \mathbb{R}^{6\times 6}$ be an orthogonal projection of rank three. Put*

$$R = \sum_{i=1}^{6}\left(P_{ii} - \frac{1}{2}\right)^2.$$

If $R \ge 8/27$, then there is a three-element set I such that $P_{I,I} \succeq I_3/6$.

Proof. By Proposition 5.4, we may assume

$$\left|P_{ii} - \frac{1}{2}\right| < \frac{13}{36} \qquad (1 \le i \le 6). \tag{35}$$

Partition the twenty three-element sets into ten complementary pairs. In each pair choose a member I for which

$$x_I := \operatorname{tr} P_{I,I} - \frac{3}{2} \ge 0,$$

and put

$$d_I = \det P_{I,I}, \qquad d_I^c = \det P_{I^c,I^c}.$$

The eigenvalues of the two complementary compressions are $\lambda_1, \lambda_2, \lambda_3$ and $1-\lambda_1, 1-\lambda_2, 1-\lambda_3$. It follows that

$$\det\left(P_{I,I} - \frac{1}{6}I_3\right) = \frac{5}{6}d_I - \frac{1}{6}d_I^c - \frac{5}{36}x_I - \frac{5}{108}. \tag{36}$$

For a high-trace member, nonnegativity of the determinant in (36) is equivalent to $P_{I,I} \succeq I_3/6$. Indeed, if two eigenvalues were below $1/6$, then $0 \preceq P_{I,I} \preceq I_3$ would give $\operatorname{tr} P_{I,I} < 4/3$, contrary to the choice of I; the case of a zero shifted eigenvalue follows by the same argument.

Suppose that every high-trace member fails. Equation (36) then gives

$$5d_I - d_I^c < \frac{5}{18} + \frac{5}{6}x_I.$$

The determinantal marginal identity $\sum_{J\ni i} d_J = P_{ii}$ implies

$$\sum_I x_I(d_I - d_I^c) = \sum_{|J|=3} d_J\left(\operatorname{tr} P_{J,J} - \frac{3}{2}\right) = R.$$

Moreover, direct counting gives
$$\sum_I x_I^2 = 3R.$$
Multiplying the preceding inequality by x_I, summing over the ten pairs, and using
$$5d_I - d_I^c = 2(d_I + d_I^c) + 3(d_I - d_I^c)$$
yields
$$2\sum_I x_I(d_I + d_I^c) + \frac{1}{6}\sum_I x_I^2 < \frac{5}{18}\sum_I x_I. \tag{37}$$
The first sum in (37) is at least R, because
$$\sum_I x_I(d_I + d_I^c) = \sum_{|J|=3} d_J \left|\operatorname{tr} P_{J,J} - \frac{3}{2}\right| \geq \left|\sum_{|J|=3} d_J \left(\operatorname{tr} P_{J,J} - \frac{3}{2}\right)\right| = R.$$
Thus, writing $S = \sum_I x_I$, the preceding square identity and (37) imply
$$9R < S.$$
To sharpen the direct Cauchy–Schwarz estimate, put $z_i = P_{ii} - 1/2$ and represent the ten pairs by the triples containing coordinate 6. Then
$$S = \sum_{1\leq i<j\leq 5} |z_6 + z_i + z_j|.$$
For any choice of signs in this absolute-value sum, regard the negative signs as the edges of a graph on five vertices. If the graph has m edges and degrees $e_1, \ldots, e_5$, the coefficient vector, after restriction to $\sum_i z_i = 0$, is
$$c_6 = 5 - m, \qquad c_i = m - 1 - 2e_i \quad (1 \leq i \leq 5).$$
Lemma 5.5 gives the maximal squared norms
$$30, \quad 24, \quad 22, \quad 24, \quad 30, \quad 24.$$
The cases $m > 5$ follow by reversing all signs. The two cases with norm squared 30 are precisely the permutations and sign changes of $(-5, 1, 1, 1, 1, 1)$; their value on z is $6|z_i| < 13/6$ by (35). Every other sign vector has norm at most $\sqrt{24}$. Consequently
$$S \leq \max\left\{\frac{13}{6}, \sqrt{24R}\right\}.$$
If $R \geq 8/27$, then both $13/6 < 9R$ and $\sqrt{24R} \leq 9R$, contradicting $9R < S$. □

Proof of Theorem 1.7. Put $P = UU^T$, and let $0 < r_1 \leq r_2 \leq r_3 \leq 1$ be the roots of the mixed polynomial (17). The proof of Proposition 2.14 selects a three-set I with $\lambda_{\min}(P_{I,I}) \geq r_1$, and the same theorem gives $r_1 \geq 1/7$.

Let α_6 be the smallest zero of
$$g(x) = 70x^3 - 120x^2 + 57x - 6.$$
Since
$$g\left(\frac{1}{7}\right) = -\frac{5}{49} < 0, \qquad g\left(\frac{3}{20}\right) = \frac{69}{800} > 0,$$

and g is increasing on $[1/7, 3/20]$, one has

$$\frac{1}{7} < \alpha_6 < \frac{3}{20}. \tag{38}$$

We prove that the mixed polynomial

$$\chi(x) = x^3 - Qx^2 + Bx - C$$

satisfies $\chi(\alpha_6) \leq 0$. Lemma 2.15 and Corollary 2.11 give

$$Q - \frac{9}{10} \leq B, \qquad B \leq 2C + \frac{2}{3}Q - \frac{1}{2}, \qquad B \leq \frac{15}{2}C + \frac{Q}{10}. \tag{39}$$

Also $Q \geq 3/2$ by Cauchy–Schwarz. Fix Q temporarily and maximize

$$\alpha_6^3 - Q\alpha_6^2 + \alpha_6 B - C$$

over B, C satisfying (39). The two upper bounds for B meet at

$$C_0(Q) = \frac{17Q - 15}{165}, \qquad B_0(Q) = \frac{3(32Q - 25)}{110}.$$

Because $15\alpha_6/2 - 1 > 0$ and $2\alpha_6 - 1 < 0$, the maximum occurs at this intersection whenever it is compatible with the lower bound for B. The compatibility condition is

$$B_0(Q) - Q + \frac{9}{10} = \frac{12 - 7Q}{55} \geq 0,$$

or $Q \leq 12/7$. In this range the maximal value is

$$\alpha_6^3 - Q\alpha_6^2 + \alpha_6 B_0(Q) - C_0(Q).$$

Its coefficient of Q is

$$-\frac{165\alpha_6^2 - 144\alpha_6 + 17}{165} > 0.$$

The sign follows from (38); the quadratic in the numerator is negative on that interval. Hence the preceding affine expression is at most its value at $Q = 12/7$.

If $Q \geq 12/7$, feasibility forces the second upper bound in (39) to meet the lower bound first. The maximizing choice is therefore

$$B = Q - \frac{9}{10}, \qquad C = \frac{Q}{6} - \frac{1}{5}.$$

The resulting value of $\chi(\alpha_6)$ is affine in Q, with coefficient

$$-\alpha_6^2 + \alpha_6 - \frac{1}{6} < 0,$$

so it too is at most its value at $Q = 12/7$. At the common endpoint,

$$Q = \frac{12}{7}, \qquad B = \frac{57}{70}, \qquad C = \frac{3}{35},$$

and consequently

$$\chi(\alpha_6) = \frac{1}{70} g(\alpha_6) = 0.$$

Thus $\chi(\alpha_6) \leq 0$. If $r_1 < \alpha_6$, then

$$r_2 \geq Q - r_3 - r_1 \geq \frac{1}{2} - r_1 > \frac{1}{2} - \alpha_6 > \alpha_6,$$

where $Q \geq 3/2$, $r_3 \leq 1$, and (38) were used. The sign pattern of a monic cubic would then give $\chi(\alpha_6) > 0$, a contradiction. Hence $r_1 \geq \alpha_6$, and the interlacing selection proves the theorem. $\square$

Remark 5.7. The bound in Theorem 1.7 is stronger than the bound supplied by Theorem 1.1 at $n = 6$. Let $g(x) = 70x^3 - 120x^2 + 57x - 6$. Then

$$g\left(\frac{183}{1250}\right) = -\frac{14668410}{1250^3} < 0,$$

so $\alpha_6 > 183/1250$. On the other hand,

$$\tau_6 = \frac{118 - \sqrt{5412}}{304} < \frac{183}{1250},$$

since

$$5412 \cdot 1250^2 - (118 \cdot 1250 - 304 \cdot 183)^2 = 16520576 > 0.$$

Thus $\alpha_6 > \tau_6$.

The linear-programming calculation in the proof identifies the endpoint $Q = 12/7$ of the feasible region for the aggregate quantities Q, B, C. The diagonal restrictions in Theorem 1.8 come instead from Propositions 5.4 and 5.6. The two arguments serve different purposes: the former improves the spectral lower bound, whereas the latter restricts the geometry of a possible counterexample.

Proof of Theorem 1.8. Suppose that (4) holds. If the nonzero rows of U belonged to at most five lines, Theorem 1.3 would give a three-element set I with $\sigma_{\min}(U_I)^2 \geq 1/6$, a contradiction. Thus all six rows are nonzero and determine six distinct lines.

Proposition 5.3 shows that the row matroid has two disjoint bases. Since the ground set has six elements, these bases have the form I and I^c for some three-element set I; equivalently, both U_I and U_{I^c} are nonsingular.

By Proposition 5.4,

$$P_{ii} < \frac{31}{36} \qquad (1 \leq i \leq 6).$$

The matrix $I_6 - P$ is also a rank-three orthogonal projection. Its three-by-three principal submatrices satisfy

$$\lambda_{\min}((I_6 - P)_{I,I}) = 1 - \lambda_{\max}(P_{I,I}) = \lambda_{\min}(P_{I^c,I^c}),$$

where the last equality follows from the spectral identity used in the proof of Proposition 5.4. Hence $I_6 - P$ also satisfies the counterexample assumption. Applying the same upper bound to $I_6 - P$ gives $1 - P_{ii} < 31/36$, and therefore $P_{ii} > 5/36$ for every i.

Proposition 5.6 gives

$$\sum_{i=1}^{6} \left(P_{ii} - \frac{1}{2}\right)^2 < \frac{8}{27}. \tag{40}$$

Finally, Theorem 1.7 and (4) yield

$$\alpha_6 \leq \max_{|I|=3} \sigma_{\min}(U_I)^2 < \frac{1}{6}. \tag{41}$$

All assertions follow. □

6 Conclusion and Open Problems

For arbitrary $n \geq 6$, the mixed characteristic polynomial and the inequalities for Q, B, C give the lower bound $\Lambda_n \geq \tau_n$ and the asymptotic estimate

$$\liminf_{n\to\infty} n\Lambda_n \geq \frac{5 - \sqrt{13}}{2}.$$

When the nonzero rows occupy at most five lines, the bound $n^{-1/2}$ holds; in particular, $\Lambda_5 = 1/5$. The construction in Theorem 1.5 shows that $n^{-1/2}$ is optimal for every $n > k$. In the first unresolved self-dual case $(n, k) = (6, 3)$, Theorem 1.7 improves the general bound to α_6, and Theorem 1.8 confines every possible counterexample to a bounded region.

The remaining problem is to exclude a rank-three projection $P \in \mathbb{R}^{6\times 6}$ satisfying all the conditions in Theorem 1.8 and

$$\lambda_{\min}(P_{I,I}) < \frac{1}{6} \qquad \text{for every three-element set } I.$$

The aggregate inequalities for Q, B, C do not retain all incidence information among the twenty three-element sets. One possible final step is a finite semialgebraic decision problem. Write a symmetric 6×6 matrix as $(p_{ij})_{1\le i\le j\le 6}$. The projection condition $P^2 = P$ gives twenty-one quadratic equations in these twenty-one variables; the equation $\operatorname{tr} P = 3$ fixes the rank. Add the inequalities in Theorem 1.8. For each three-element set I, the condition

$$\lambda_{\min}(P_{I,I}) < \frac{1}{6}$$

is semialgebraic; for example, it can be written by introducing a unit vector $v_I \in \mathbb{R}^3$ and requiring

$$v_I^T \left(P_{I,I} - \frac{1}{6} I_3 \right) v_I < 0.$$

The ten complementary pairs may be separated according to the signs of $\operatorname{tr} P_{I,I} - 3/2$. Without an argument excluding equality, this gives at most 3^{10} sign patterns, not 2^{10}; the action of S_6 identifies many of them. Each remaining branch is a finite semialgebraic system and can, in principle, be decided by quantifier elimination or cylindrical algebraic decomposition. The strict inequalities should be retained in this formulation; compactness of the projection variety alone does not replace the emptiness test.

References

[1] S. A. Goreinov, E. E. Tyrtyshnikov, and N. L. Zamarashkin, A theory of pseudoskeleton approximations, *Linear Algebra Appl.* **261** (1997), 1–21. `https://doi.org/10.1016/S0024-3795(96)00301-1`.

[2] J. Edmonds, Minimum partition of a matroid into independent subsets, *J. Res. Nat. Bur. Standards Sect. B* **69B** (1965), 67–72. `https://nvlpubs.nist.gov/nistpubs/jres/69B/jresv69Bn1-2p67_A1b.pdf`.

[3] Y. Nesterenko, Submatrices with the best-bounded inverses: revisiting the hypothesis, arXiv:2303.07492, 2023. `https://arxiv.org/abs/2303.07492`.

[4] Y. Nesterenko, Submatrices with the best-bounded inverses: studying $\mathbb{R}^{n\times 2}$ and $\mathbb{C}^{n\times 2}$, arXiv:2408.16631, 2024. `https://arxiv.org/abs/2408.16631`.

[5] R. Sengupta and M. Pautov, On the submatrices with the best-bounded inverses, arXiv:2604.05944, 2026. `https://arxiv.org/abs/2604.05944`.